



SpearDAT

A Modern C++ and MATLAB Interface for Missile DATCOM
Aerodynamic Methods

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The Starting Point: Missile DATCOM

AFRL-VA-WP-TR-1998-3009
This Report Supersedes WL-TR-93-3043



MISSILE DATCOM
USERS MANUAL - 1997 FORTRAN 90 REVISION

William B. Blake
Air Force Research Laboratory
Air Vehicles Directorate
Wright-Patterson Air Force Base, Ohio 45433-7531

FEBRUARY 1998
FINAL REPORT FOR PERIOD APRIL 1993-DECEMBER 1997

APPROVED FOR PUBLIC RELEASE; DISTRIBUTION IS UNLIMITED

AIR VEHICLES DIRECTORATE
AIR FORCE RESEARCH LABORATORY
AIR FORCE MATERIEL COMMAND
WRIGHT PATTERSON AIR FORCE BASE, OHIO 45433-7562

AFWAL-TR-86-3091

DTIC # AD-A211 086

MISSILE DATCOM
VOLUME I - FINAL REPORT

STEVEN R. VUKELICH
STAN L. STOY
KEITH A. BURNS
JOSEPH A. CASTILLO
MARVIN E. MOORE

McDonnell Douglas Missile Systems Company
St. Louis, Missouri 63133

DECEMBER 1988
FINAL REPORT FOR PERIOD SEPTEMBER 1981- DECEMBER 1985

Approved for public release; distribution unlimited

FLIGHT DYNAMICS LABORATORY
AIR FORCE WRIGHT AERONAUTICAL LABORATORIES
AIR FORCE SYSTEMS COMMAND
WRIGHT-PATTERSON AIR FORCE BASE, OHIO
45433-6553

- **Preliminary-design aerodynamics**

Developed to rapidly predict aerodynamic characteristics of conventional missile configurations

- **Component build-up methodology**

Bodies, lifting surfaces and their interaction are assembled using theoretical and semi-empirical prediction methods.

- **Broad engineering capability**

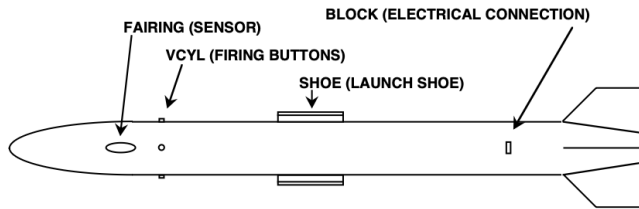
Flight-condition schedules, multiple fin sets, controls and trim, dynamic derivatives, inlets, pressure distribution estimation and experimental-data substitution evolved through successive releases.

- **Public technical foundation**

Detailed methodology, user documentation and source material provide traceability into both aerodynamic methods and the implementation

How Missile DATCOM Is Used

A batch-oriented, text-file workflow built around NAMELIST and control-card inputs.



```
CASEID PROTUBERANCE EXAMPLE CASE
DIM IN
NO LAT
$REFQ XCG=39.0,$
$FLTCON NMACH=3.,MACH=0.4,0.8,2.0,
REN=3.E06,3.E06,3.E06,ALT=0.0,
NALPHA=5.,ALPHA=-8.,-4.,0.,4.,8.,$
$AXIBOD TNOSE=OGIVE,LNOSE=12.0,DNOSE=12.0,
LCENTR=54.0,DCENTR=12.0,
TAFT=CONE,LAFT=12.0,DAFT=6.0,DEXIT=5.0,$
$PROTUB NPROT=4.,
PTYPE=FAIRING,VCYL,SHOE,BLOCK,
XPROT=14.,22.,39.,56.,
NLOC=2.,4.,2.,1.,
LPROT=5.,1.,10.,10.,10.,0.5,
WPROT=2.,1.,4.,0.25,1.,1.,
HPROT=2.,0.5,0.1,0.75,0.25,0.25,
OPROT=0.,0.,0.,0.1,0.85,0.,$
$FINSET1 SSPAN=0.0,9.0,CHORD=14.0,8.0,
XLE=64.0,SWEEP=0.0,STA=1.0,NPANEL=4.,
PHIF=45.,135.,225.,315.,$
PRINT GEOM BODY
PRINT AERO BODY
SAVE
NEXT CASE
```

1. Prepare for005.dat



2. Run Missile DATCOM



3. Interpret output files

- for006 – report
- for003 – aerodynamics
- for009-for012 – detailed outputs

```
***** THE USAF AUTOMATED MISSILE DATCOM * REV 5/97 ***** CASE 1
AERODYNAMIC METHODS FOR MISSILE CONFIGURATIONS
PLANAR WING, CRUCIFORM PLUS TAIL CONFIGURATION
STATIC AERODYNAMICS FOR BODY-FIN SET 1 AND 2
PAGE 21

***** FLIGHT CONDITIONS AND REFERENCE QUANTITIES *****
MACH NO = 2.36 REYNOLDS NO = 3.000E+06 /FT
SIDESLIP = 0.00 DEG ROLL = 0.00 DEG
REF AREA = 11.045 IN**2 MOMENT CENTER = 18.750 IN
REF LENGTH = 3.75 IN LAT REF LENGTH = 3.75 IN

ALPHA ---- LONGITUDINAL ---- -- LATERAL DIRECTIONAL --
CN CM CA CY CLN CLL
0.00 0.000 0.000 0.466 0.000 0.000 0.000
4.00 1.179 -1.577 0.465 0.000 0.000 0.000
8.00 2.476 -3.194 0.463 0.000 0.000 0.000
12.00 3.992 -5.056 0.458 0.000 0.000 0.000
16.00 5.582 -6.962 0.453 0.000 0.000 0.000
20.00 7.151 -8.904 0.445 0.000 0.000 0.000
24.00 8.601 -10.792 0.436 0.000 0.000 0.000
28.00 10.093 -12.692 0.426 0.000 0.000 0.000

ALPHA CL CD CL/CD X-C.P.
0.00 0.000 0.466 0.000 -1.389
4.00 1.144 0.546 2.093 -1.337
8.00 2.387 0.803 2.974 -1.290
12.00 3.810 1.279 2.980 -1.266
16.00 5.241 1.974 2.658 -1.247
20.00 6.568 2.864 2.293 -1.245
24.00 7.680 3.897 1.971 -1.255
28.00 8.711 5.114 1.703 -1.258

X-C.P. MEAS. FROM MOMENT CENTER IN REF. LENGTHS, NEG. AFT OF MOMENT CENTER
***** THE USAF AUTOMATED MISSILE DATCOM * REV 5/97 ***** CASE 1
AERODYNAMIC METHODS FOR MISSILE CONFIGURATIONS
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REF AREA = 11.045 IN**2 MOMENT CENTER = 18.750 IN
REF LENGTH = 3.75 IN LAT REF LENGTH = 3.75 IN

ALPHA ---- DERIVATIVES (PER DEGREE) ----
CNA CMA CYB CLNB CLLB
0.00 0.2802 -0.3892 -0.1993 0.4893 0.0000
4.00 0.3094 -0.3992 -0.2082 0.4797 -0.0105
8.00 0.3514 -0.4345 -0.2150 0.4493 -0.0187
12.00 0.3882 -0.4711 -0.2151 0.3594 -0.0110
16.00 0.3949 -0.4810 -0.2037 0.2255 0.0080
20.00 0.3774 -0.4787 -0.1896 0.1163 0.0320
24.00 0.3676 -0.4735 -0.1740 0.0449 0.0486
28.00 0.3780 -0.4766 -0.1642 -0.0028 0.0615

PANEL DEFLECTION ANGLES (DEGREES)
SET FIN 1 FIN 2 FIN 3 FIN 4 FIN 5 FIN 6 FIN 7 FIN 8
1 0.00 0.00 0.00 0.00
2 0.00 0.00 0.00 0.00

BODY ALONE LINEAR DATA GENERATED FROM SECOND ORDER SHOCK EXPANSION METHOD
```

Engineer ↔ formatted text decks ↔ batch program ↔ output files

Why SpearDAT exists

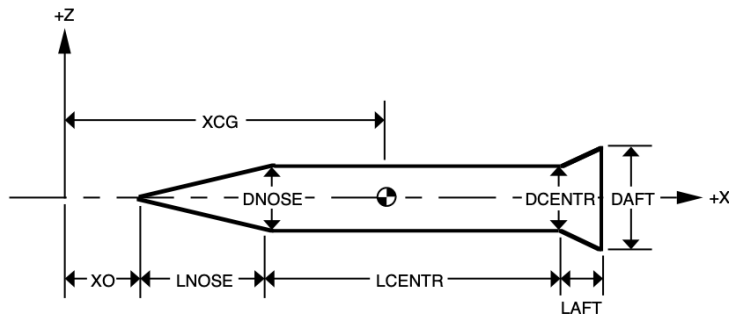
Bringing established preliminary-design aerodynamic methods into a modern engineering workflow

Historical Capability

Missile DATCOM

Preliminary design

Broad configuration capability

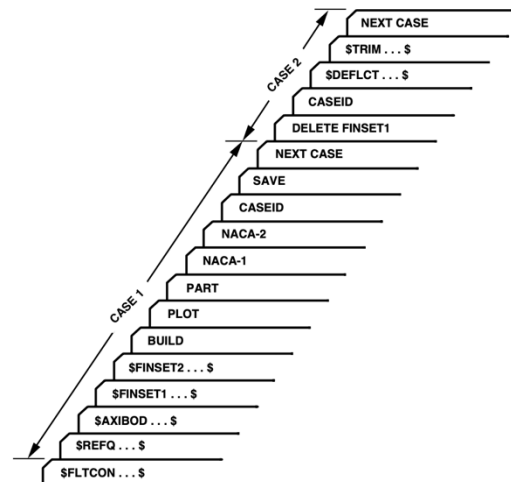


Workflow Gap

Legacy implementation

Card/file workflow

Difficult integration

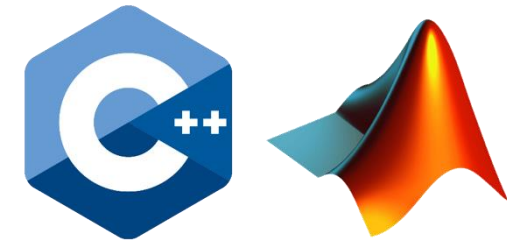


SpearDAT

Modern C++17

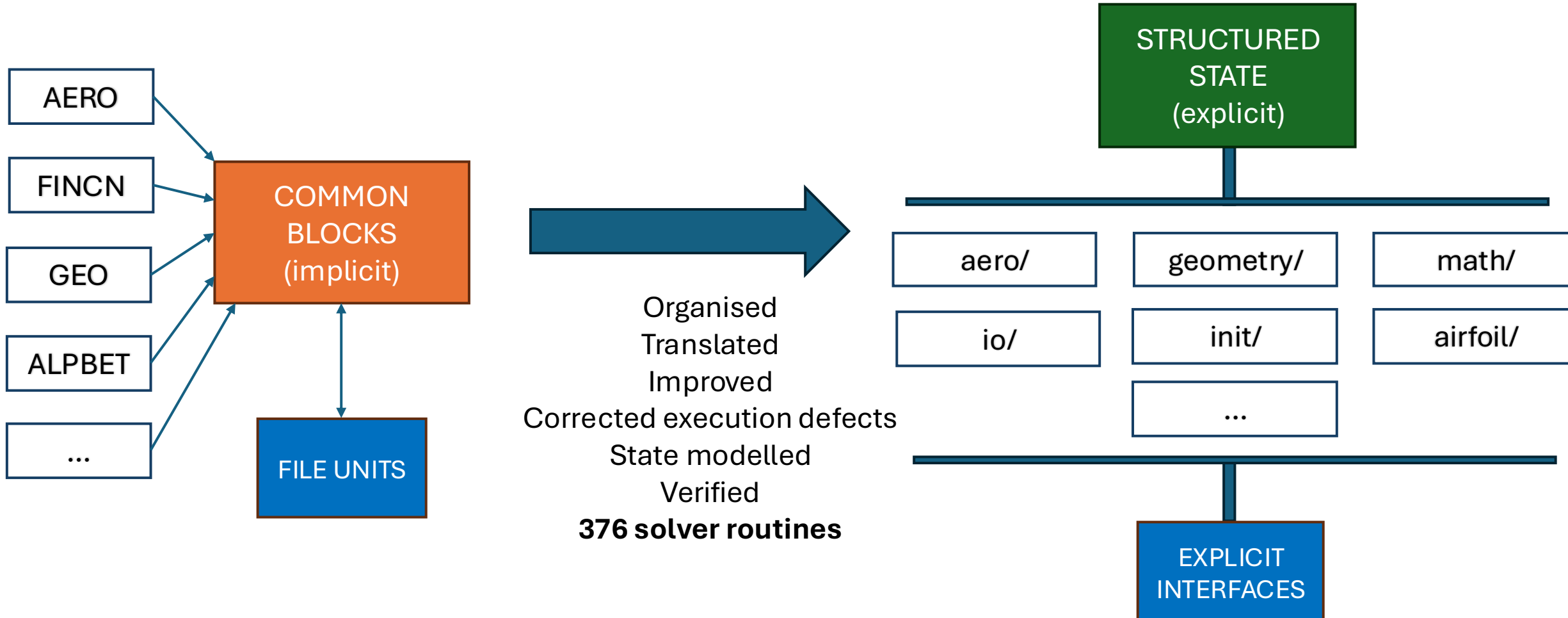
Structured API

MATLAB interface



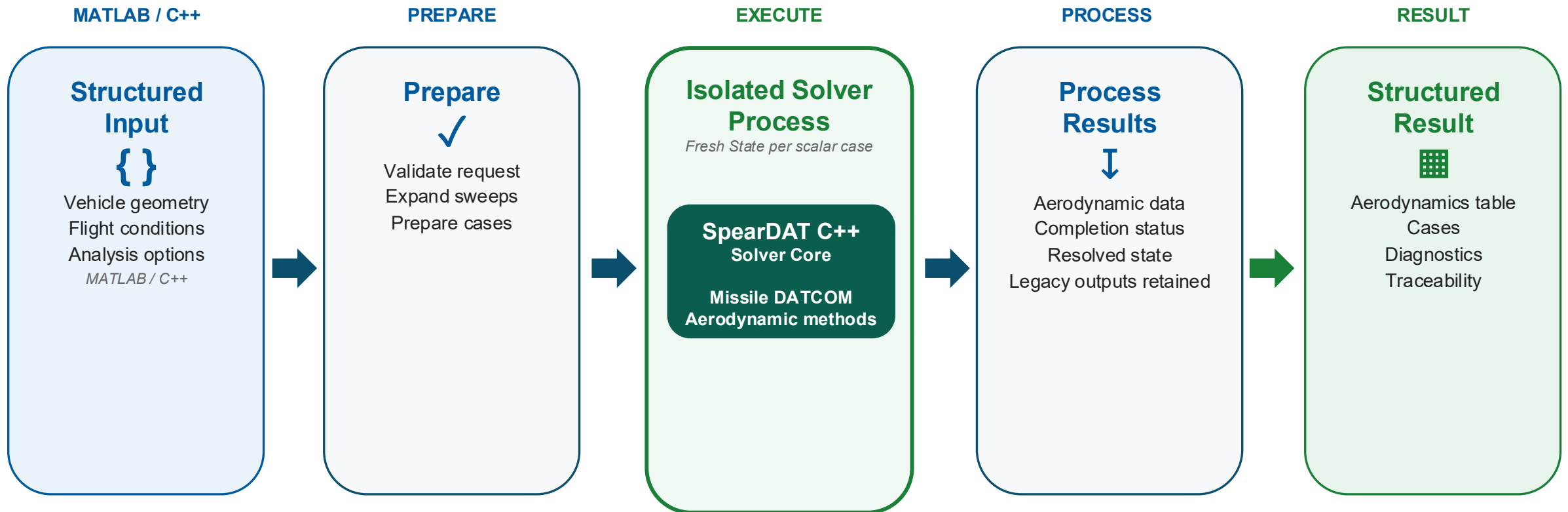
From Legacy Source to a Modern Solver

Reconstructing and modernising the publicly released implementation of the Missile DATCOM aerodynamic methods



The SpearDAT Workflow

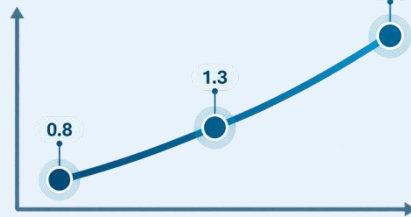
From structured engineering inputs to traceable aerodynamic results



Flight Conditions and Reference Quantities

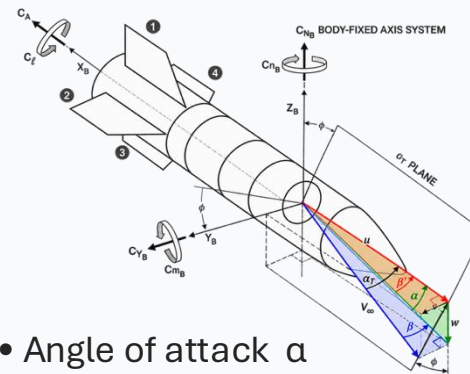
Defining the operating envelope and aerodynamic coefficient convention

Speed & Atmosphere



- Mach schedules
- Velocity schedules
- Standard atmosphere
- Reynolds-specified
- Wind-tunnel conditions

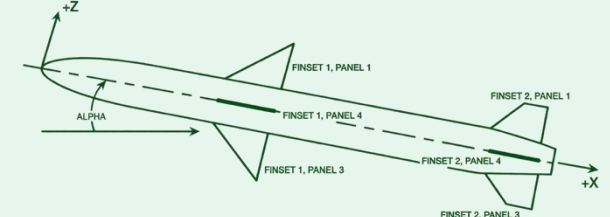
Incidence



- Angle of attack α
- Sideslip β
- Aerodynamic roll ϕ

body α/β or total- α /roll attitude

Reference Convention



- Sref
- Lref, lateral reference
- Moment reference x_{cg} , z_{cg}
- Boundary layer / roughness
- Geometric scaling

Body Geometry

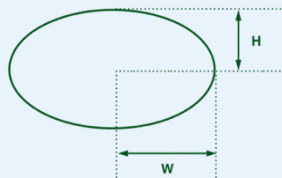
Analytic or user-defined mould lines for axisymmetric and elliptical bodies

Choose the representation

CROSS-SECTION FAMILY



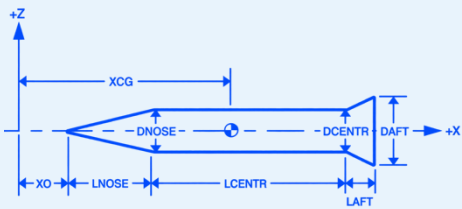
Axisymmetric
circular section



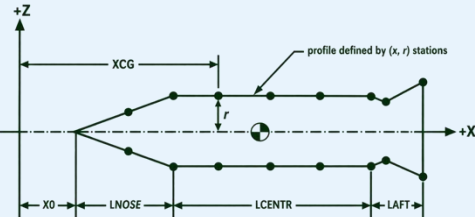
Elliptical
variable ellipticity
 $e = \text{height} / \text{width}$

+

LONGITUDINAL GEOMETRY

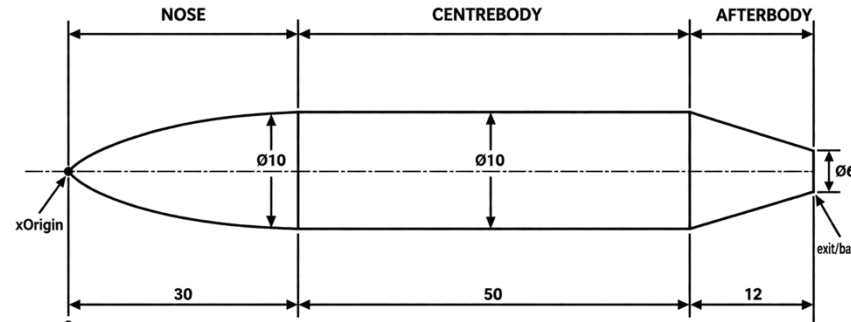


Analytic
named sections
nose → centre → aft



Stations
supplied mould line
 (x, r) or (x, w, h)

Example: analytic axisymmetric body



```
input.caseInput.body.shape = "Axisymmetric";  
input.caseInput.body.axisymmetric.geometryMode = "Analytic";  
input.caseInput.body.axisymmetric.nose.type = "Ogive";  
input.caseInput.body.axisymmetric.nose.length = 30;  
input.caseInput.body.axisymmetric.nose.diameter = 10;  
input.caseInput.body.axisymmetric.center.length = 50;  
input.caseInput.body.axisymmetric.center.diameter = 10;  
input.caseInput.body.axisymmetric.aft.type = "Conical";  
input.caseInput.body.axisymmetric.aft.length = 12;  
input.caseInput.body.axisymmetric.aft.diameter = 6;
```

Analytic noses:

- Conical
- Ogive
- Power
- Haack
- Von Karman

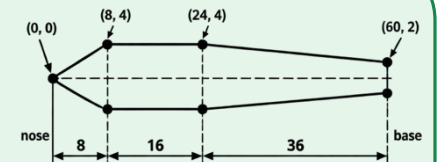
Afterbodies:

- Conical
- Ogive

Nose end:

- Sharp
- Blunted
- Truncated

Station mode



```
stations.x = [0 8 24 60];  
stations.radius = [0 4 4 2];
```

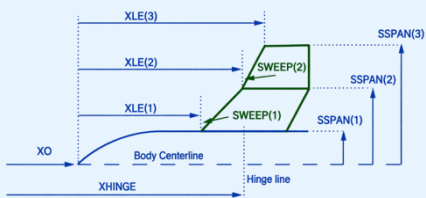
Lifting Surfaces & Controls

Fin sets define canards, wings, strakes and tails through planform, section and panel placement

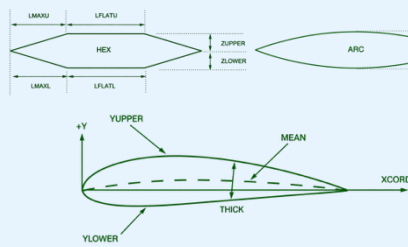


- Up to 4 fin sets • 1–8 panels/set • Up to 10 span stations/set

PLANFORM



SECTION



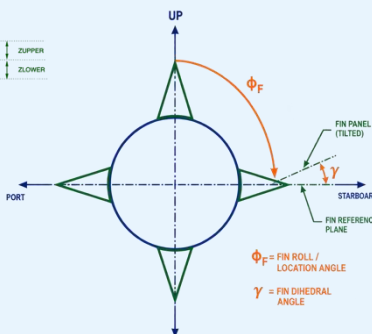
HEX

ARC

USER

NACA

PANELS



panel roll + dihedral

```
wing.set = 1;
wing.spanStations = [0 2 6 8];
wing.chord = [4 4 3 3];
wing.xLeadingEdge = [10 10.73 12.03 12.03];
wing.flapChordRatio = [0 .25 .25 0];
wing.nPanels = 2;
wing.panelRollDeg = [90 270];
wing.dihedralDeg = [-5 5];
wing.section.type = "Naca";
wing.section.naca.sectionType = "FourDigit";
wing.section.naca.designation = "0012";
```

```
tail = struct;
tail.set = 2;
tail.spanStations = [0 2.0];
tail.chord = [2.5 1.0];
tail.xLeadingEdge = [28.0 29.0];
tail.flapChordRatio = [0 0];
tail.nPanels = 4;
tail.panelRollDeg = [0 90 180 270];
tail.dihedralDeg = [0 0 0 0];
tail.section.type = "User";
tail.section.user.xChord = ...
[0 0.25 0.5 0.75 1.0];
tail.section.user.mean = [0 0 0 0 0];
tail.section.user.thickness = ...
[0 0.08 0.10 0.06 0];
```

```
input.caseInput.finSets = [wing tail];
```

Also supported:

- Fixed deflection
- Hinge-line definition
- Bending moments
- Static trim

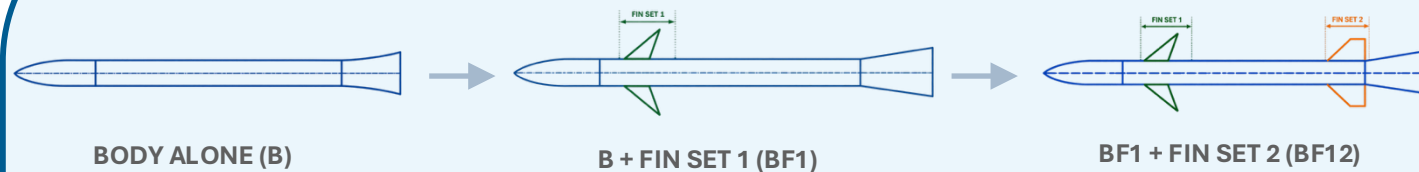
Deflection

```
defl = struct;
defl.set = 1;
defl.panelDeflectionDeg = [5 -5];
input.caseInput.deflections.finSets =
defl;
```

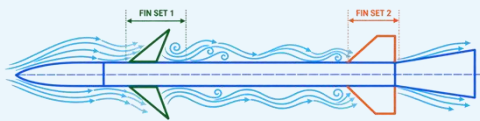
Configuration Build-up & Aerodynamic Interference

The complete vehicle is not the arithmetic sum of isolated components

Cumulative configuration synthesis

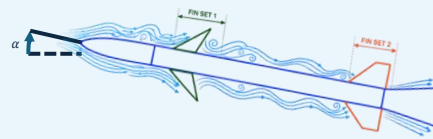


Body-fin and fin-body carryover factors



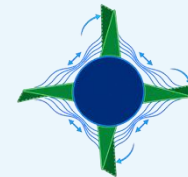
Body changes the local flow seen by the fin
Fin loading produces an aerodynamic contribution on the body

Vortex influences



body vortex $\rightarrow$ fins
forward fins $\rightarrow$ tail
changes local incidence

Panel/deflection interaction



Adjacent panels are not always aerodynamically independent

Synthesis element

Method source / reference

Body-fin upwash

Development of an Improved Carryover Interference Method for Missile DATCOM, AIAA 96-3395.

Fin-body carryover

Pitts, Nielsen & Kaattari, Lift and Center of Pressure of Wing-Body-Tail Combinations at Subsonic, Transonic, and Supersonic Speeds, NACA TR-1307, 1957; AIAA Journal, May 1981.

Carryover centre of pressure

NACA TR-1307; AIAA 94-0027; AIAA Journal, Aug. 1982.

Panel deflection interference

AGARD-R-711.

Equivalent angle of attack

Hensch & Nielsen, Equivalent Angle-of-Attack Method for Estimating Nonlinear Aerodynamics of Missile Fins, Journal of Spacecraft and Rockets, Vol. 20, No. 4, 1983.

Body-vortex strength & track

Hensch, Nielsen & Dillenius, Method for Calculating Induced Rolling Moments for Cruciform Canard Missiles at Angles of Attack up to 20 Degrees, NWC TP-5761, 1975.

Fin-vortex strength

Pitts, Nielsen & Kaattari, NACA TR-1307, 1957.

Fin-vortex track

Missile DATCOM synthesis treatment; vortex tracked downstream along the local velocity vector.

Integration note

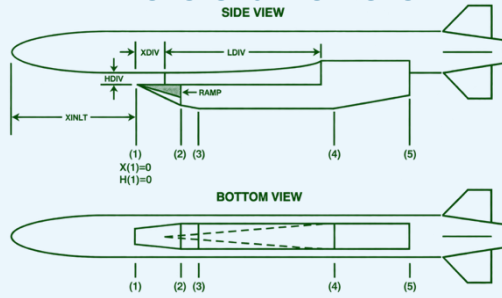
The Missile DATCOM Final Report, Vol. I documents the configuration-synthesis methodology and the rationale for selecting and integrating these component methods.

Interference can change both the magnitude and the sign of a control response.

Optional & Advanced Capabilities

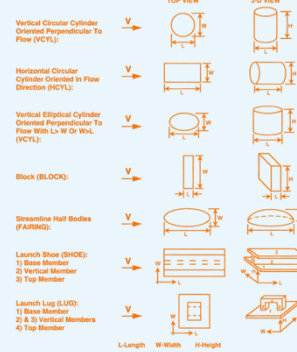
Specialised models and outputs beyond the core body-fin analysis

Inlets & diverters



installed force & moment increments
additive / spillage drag
2-D top • 2-D side • axisymmetric

Body protuberances



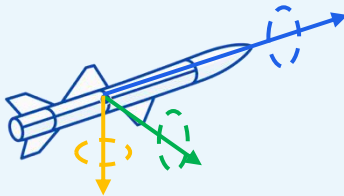
non-lifting external features
drag + local interference effects
cylinders • blocks • fairings • lugs

Experimental data substitution



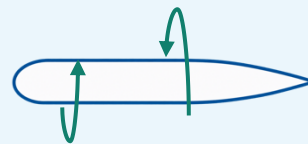
test / CFD at selected build-up stage
hybrid experimental + theoretical build-up

Dynamic derivatives



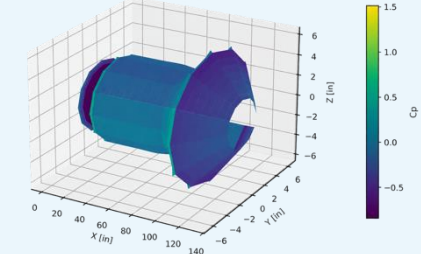
first-order response to
pitch • yaw • roll • changing incidence

Spin / Magnus



spin roll damping
spin-induced side force & yawing moment
specialised body correlation

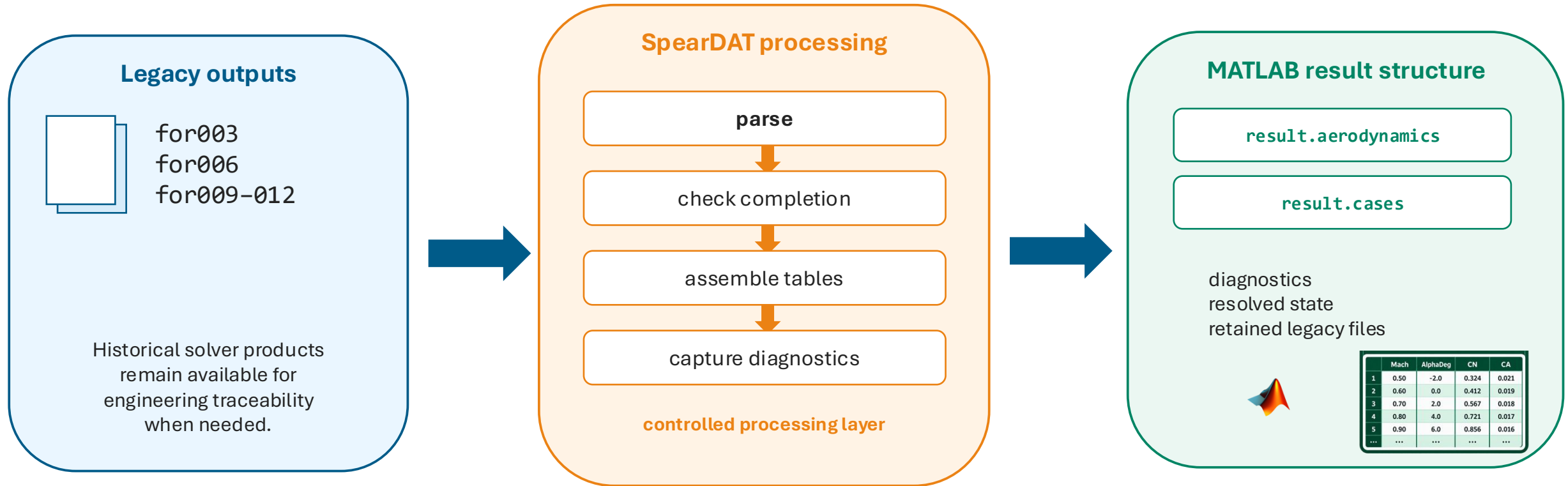
Pressure distributions



Body C_p (shown) - fin C_p also available
local Mach auxiliary outputs
method-dependent distributions

Structured Results & Traceability

From legacy solver products to MATLAB-ready aerodynamic data



Traceability retained: generated deck • case status • row ranges • diagnostics • optional legacy outputs

Verification & Validation

Different evidence answers different engineering questions

Implementation Verification

Does the software perform the intended calculation?

- routine-level regression tests
- solver-path / integration tests
- generated-deck full-stack tests
- historical Fortran comparisons where meaningful
- corrected defects tested explicitly

477 / 477 passing

native automated tests

Interface & Traceability

Did the solver receive what the user requested?

- input validation & defaults
- deterministic for005 generation
- sweep expansion into scalar cases
- resolved-state capture
- case status / diagnostics / row mapping
- retained legacy artefacts when required

MATLAB request
↓
generated deck
↓
resolved state
↓
result

TRACEABLE

Aerodynamic Validation

Do the predictions reproduce measured physical behaviour?

4 independent NASA wind-tunnel datasets

canard + tail

body + cruciform tail

elliptical body

twin inlet + tri-tail

forces • moments • controls • roll •
configuration effects

**agreement + reduced-fidelity
regions
both reported**

Passing software tests does not establish aerodynamic accuracy — both questions are assessed separately.

Experimental Validation I: NASA TM-2005-213541

Axisymmetric missile concept with controlled roll and tail deflections

Aerodynamics of an Axisymmetric Missile Concept Having Cruciform Strakes and In-Line Tail Fins From Mach 0.60 to 4.63

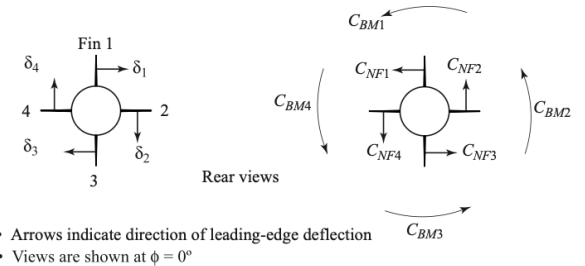
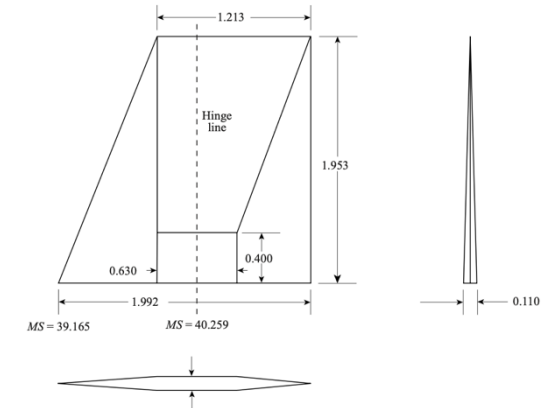
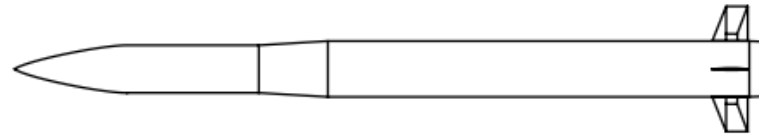
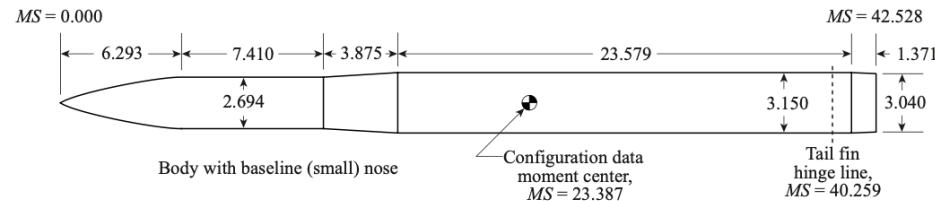
Jerry M. Allen, NASA/TM-2005-213541,
March 2005

NASA/TM-2005-213541



Aerodynamics of an Axisymmetric Missile Concept Having Cruciform Strakes and In-Line Tail Fins From Mach 0.60 to 4.63

Jerry M. Allen
Langley Research Center, Hampton, Virginia



Test Envelope

- Mach: 0.60 to 4.63
- Alpha: -2° to 22°
- Roll angle: -90° to 90°
- Tail fin deflections: varied

Selected validation case

- Configuration 3
- Body + cruciform tail
- No strakes
- Controlled roll/control-surface deflections

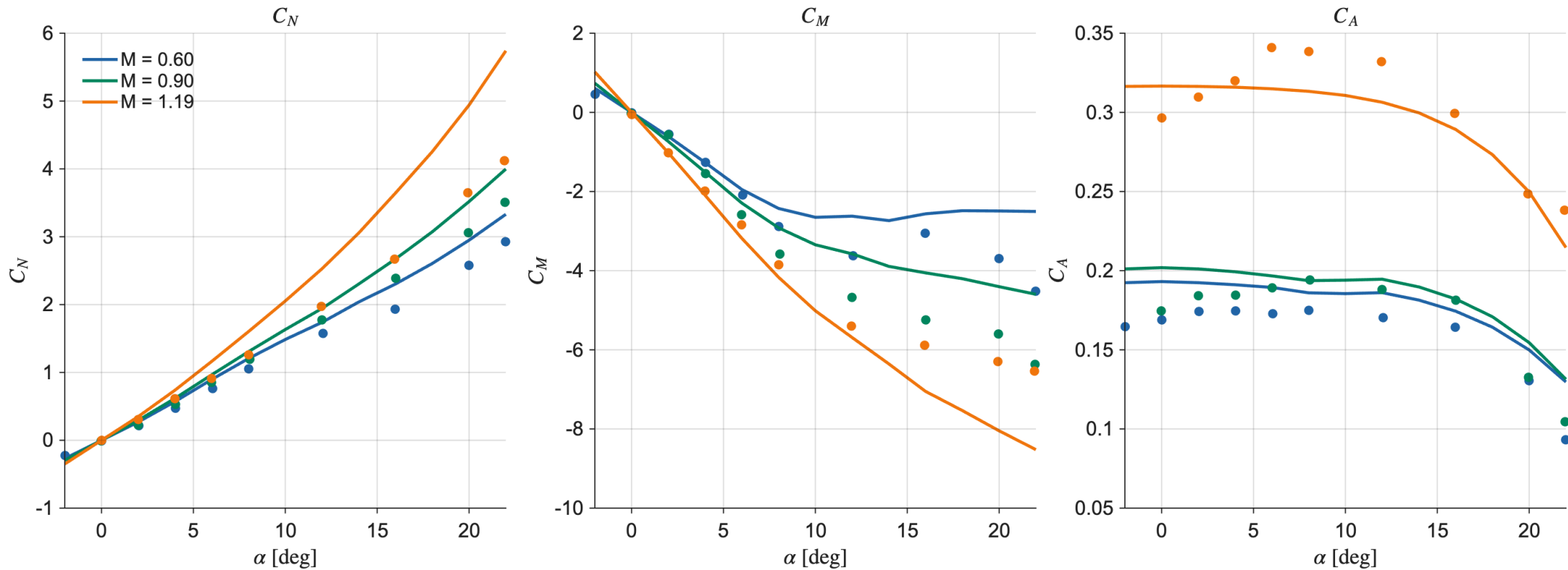
Why this dataset?

- Electronic wind-tunnel dataset
- Many runs across Mach/alpha/roll/control inputs
- Exercises longitudinal, lateral and control-response behaviour

NASA 2005: Longitudinal Correlation

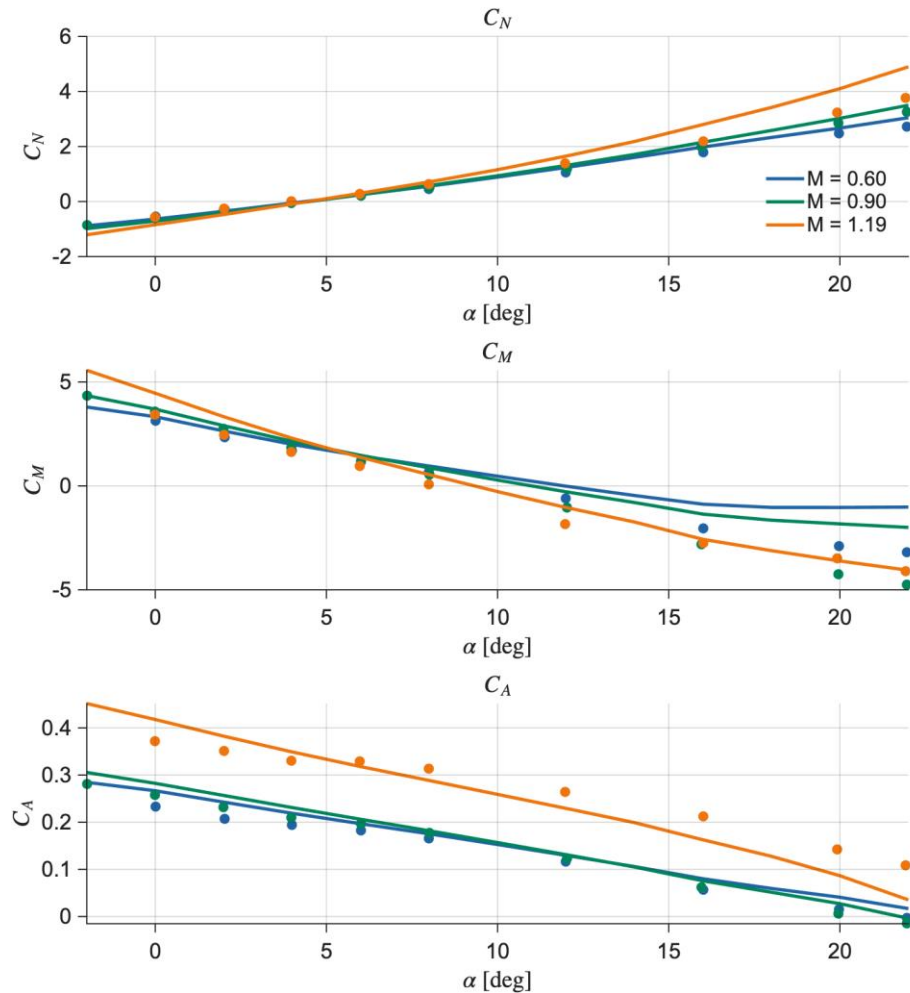
NASA 2005 Configuration 3: baseline PHI = 0 correlation

Colour = Mach number | Solid lines: SpearDAT | Filled circles: NASA experiment

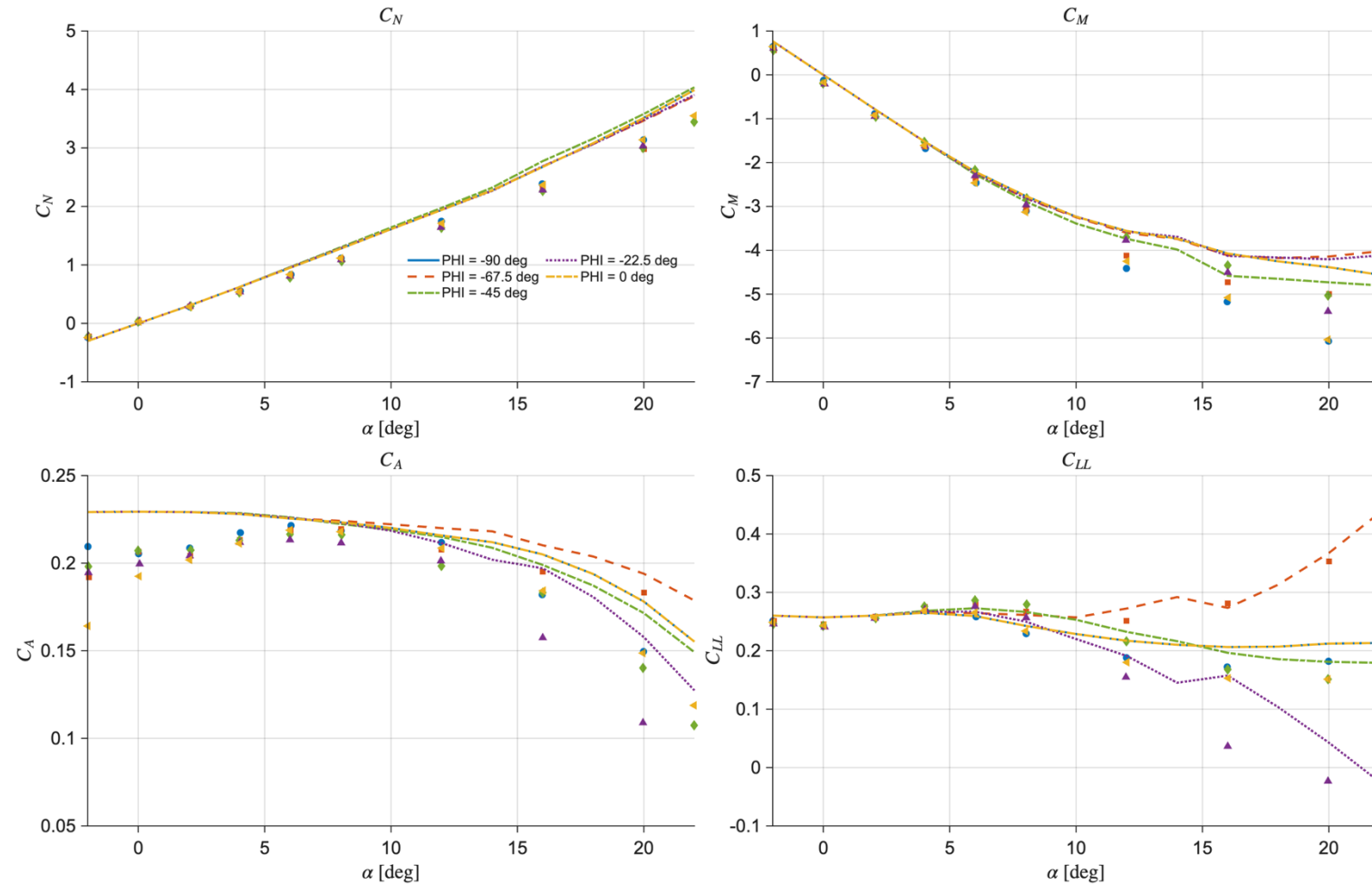


NASA 2005: Roll & Control

NASA 2005 Configuration 3: physical Pitch +10 deg
 PHI = 0 deg | Solid lines: SpearDAT | Filled circles: NASA experiment



NASA 2005 Configuration 3: physical Roll +5 deg
 M = 0.90 | Colour = body roll angle PHI | Solid lines: SpearDAT | Filled markers: NASA experiment

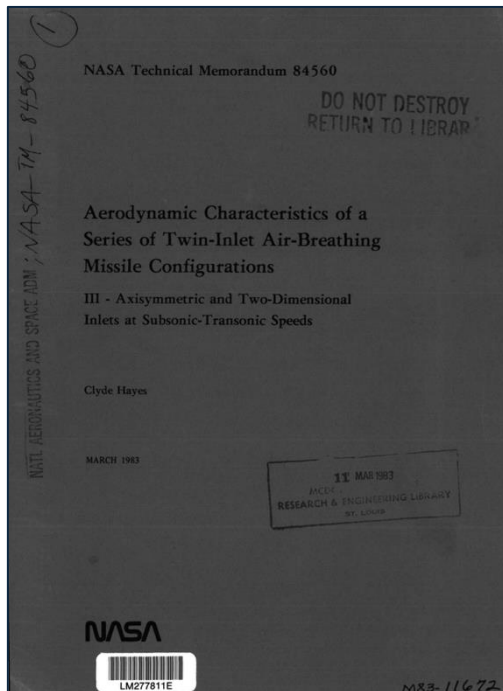
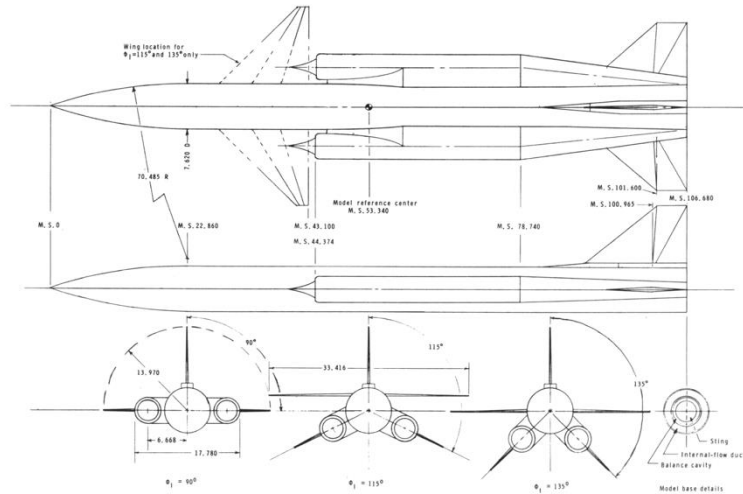


Experimental Validation II: NASA TM 84560

Axisymmetric tri-trail missile concept with an air-breathing configuration

Aerodynamic Characteristics of a Series of Twin-Inlet Air-Breathing Missile Configurations.

Clyde Hayes, NASA/TM-84560, 1983



Test Conditions

- **Mach: 0.60, 0.80, 0.95**
- **Alpha: up to 20°**
- **No sideslip/roll angle**
- **Closed internal ducts**

Selected validation case

- **Configuration B1-I1-T1**
- **Body + tritail + $\Phi 1 = 90^\circ$ inlet configuration**
- **No strakes**

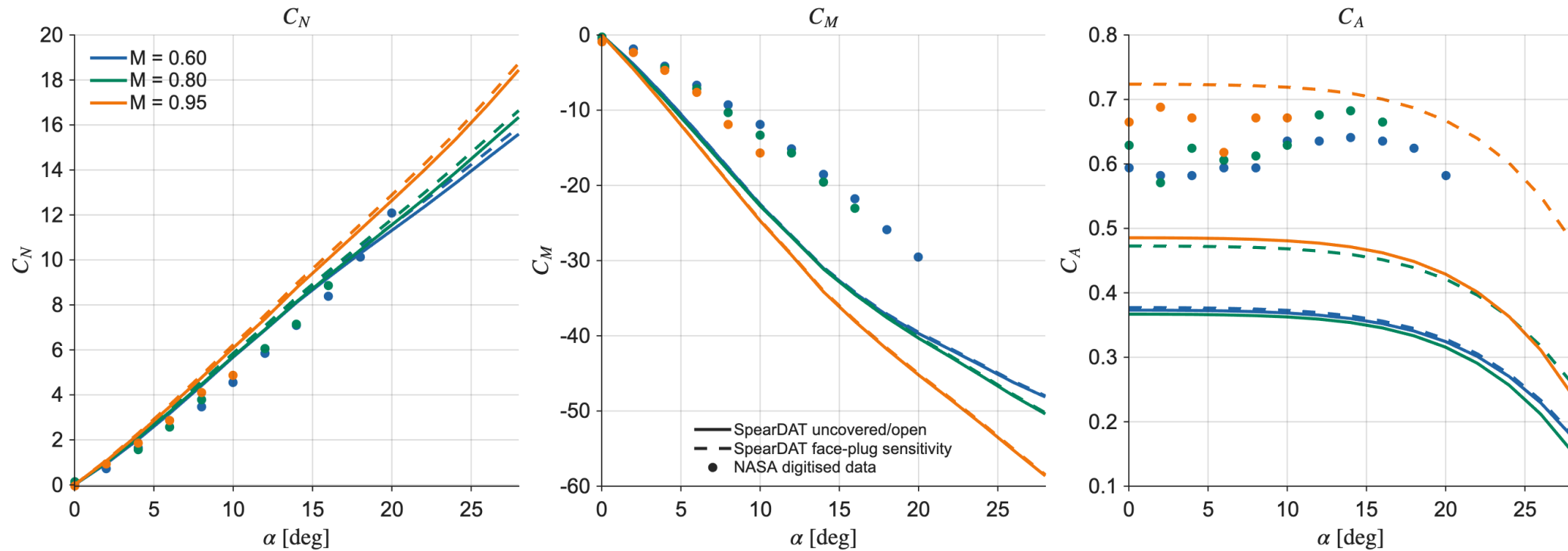
Why this dataset?

- **Dense subsonic-transonic dataset for an air-breathing missile configuration**
- **Exercises installed inlets, diverters and tri-tail build-up**
- **Useful validation case despite inlet-closure mismatch**

NASA TM 84560: Longitudinal Correlation

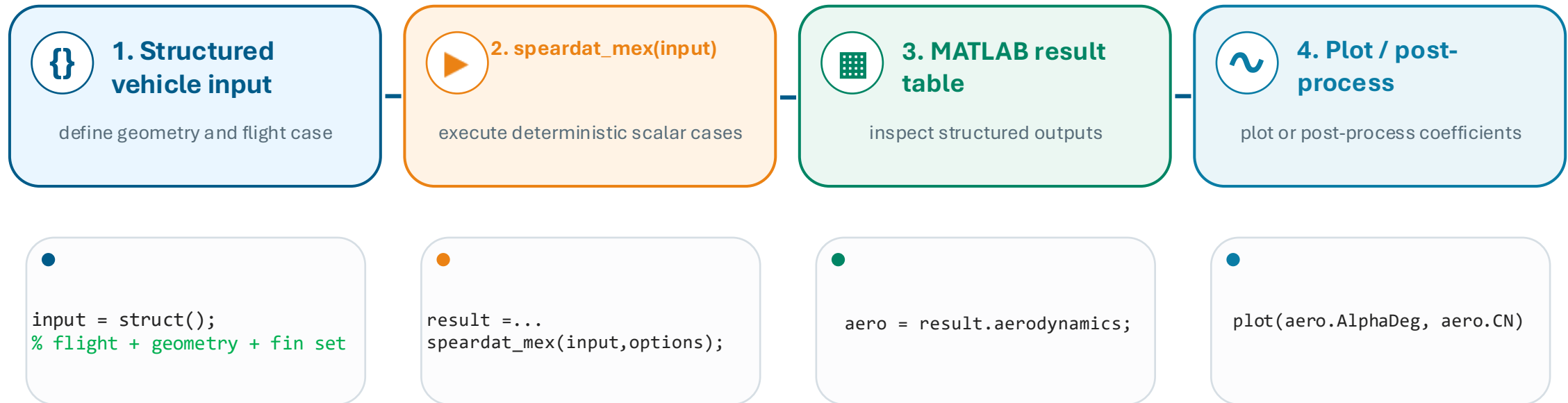
NASA TM-84560: $B_1I_1T_1$, $\phi_1 = 90^\circ$, zero tail deflection

Colour = Mach number | solid = uncovered/open | dashed = face-plug sensitivity | filled circles = NASA



Live MATLAB Demonstration

Define ➡ Run ➡ Inspect ➡ Plot



Current Status & Next Steps

A usable engineering tool today, with a clear path toward broader validation and deployment

SpearDAT today

- ✓ Reconstructed & modernised C++17 solver stack
- ✓ Typed C++ API + MATLAB MEX interface
- ✓ Structured input, validation, execution and results
- ✓ Automated verification and full-stack regression testing
- ✓ Independent experimental validation studies
- ✓ Comprehensive MATLAB Interface User Guide

376-page MATLAB Interface User Guide



Installation • quick start • capabilities •
API reference • validation • applicability



Documentation, examples and traceability
captured with the interface.

User Guide

Development roadmap

- Broader experimental validation
- Capability refinement
- Parallel execution / performance
- Qt GUI via C++ API
- Release & cross-platform packaging

Automation & design-space exploration

Parameter sweeps • optimisation • Monte Carlo • aerodynamic databases

```
fitness = @(x) run_speardat_design(x);  
[xBest,fBest] = ga(fitness,...);
```

design
vars

SpearDAT

aero
metrics

optimisation

Summary & Discussion

Modern software access to established preliminary-design aerodynamic methods

1

Modernised implementation

SpearDAT implements the publicly released Missile DATCOM aerodynamic methods in a reconstructed and modernised C++17 solver

structured state • explicit interfaces •
isolated execution • tested solver stack

2

Engineering workflow

MATLAB and C++ users work with structured inputs and structured aerodynamic results

validation • automated case preparation •
traceability • retained legacy outputs

3

Evidence-based use

Software verification and aerodynamic validation are treated as separate engineering questions

477 automated tests • independent NASA comparisons • applicability documented

Rapid preliminary-design capability in MATLAB, with the limitations of the underlying methods made explicit

Questions & Discussion

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MATLAB Interface User Guide available on request